WAITHIEGENI JEDIDAH

INDUSTRIAL CHEMIST| WATER TREATMENT DESIGN ENGINEER

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Professional summary

Industrial Chemistry graduate with practical experience in water treatment system design, laboratory analysis, materials testing and manufacturing quality control. Currently involved in designing water treatment systems based on raw water analysis and client requirements, including reverse osmosis, water softening, filtration and wastewater treatment applications.
Combines analytical laboratory skills with technical documentation, data interpretation and client communication. Interested in applying scientific and engineering principles to practical challenges in water, environmental and industrial processes.

Core Skills

- **Laboratory& Analytical Chemistry**: Analytical Chemistry| Water Analysis| Atomic Absorption Spectroscopy (AAS)| Atomic Emission Spectroscopy (AES)| Flame Photometry| Sample preparation and laboratory analysis| Laboratory reporting and data interpretation
- **Water & Environmental Engineering**: Water Treatment System Design and sizing| Raw water and Water Quality Analysis| Reverse Osmosis (RO)| Water Softening and filtration| Wastewater Treatment| Treatment process selection
- **Manufacturing & Quality Control**: Manufacturing Quality Inspections| Dimensional measurements| Tensile Strength Testing| Product Specification Verification| Quality Assurance and Documentation
- **Technical & Digital Skills**: Technical Proposals and Quotations| Technical Documentation| Laboratory Reporting| Data Interpretation
- **Tools & Instruments**: Microsoft Excel| Microsoft Word| Microsoft PowerPoint| Vernier Callipers| Micrometer Screw Gauge

Professional Experience

- **Water treatment design engineer**

Kridha Limited

May 2026 – Present

- Independently designed 15+ reverse osmosis (RO) treatment systems for borehole and municipal water applications, with design capacities ranging from 250 to 10,000 liters per hour.
- Interpreted raw water analysis results, including total dissolved solids (TDS) and constituent concentrations to establish treatment requirements and guide membrane selection.
- Selected RO membrane models and determined membrane element quantities and system configurations based on feed-water characteristics and required treatment capacity.
- Calculated feed-water and permeate flow rates and determined RO recovery percentages to support system design.
- Selected appropriate pretreatment processes and equipment, including multimedia filtration, activated carbon filtration, water softening, iron removal, cartridge filtration and chemical dosing systems.
- Identified appropriate chemical treatment requirements based on raw water characteristics and proposed treatment configurations.
- Prepared process flow diagrams, technical proposals and quotations for proposed water treatment systems.
- Collaborated with technical personnel on detailed equipment sizing and engineering drawings.
- Contributed to the design of water softening, filtration and wastewater treatment systems in collaboration with other technical personnel.
- Assesses client-specific daily water demand and reviewed mandatory raw water analysis reports to establish treatment requirements and determine appropriate RO system capacities.

- **Quality Control Assistant (***Volunteer***)**

Devki Steel Mills

February 2026 – April 2026

- Performed in-process and final quality inspections across steel manufacturing operations including acid pickling, phosphating, wire drawing and annealing
- Verified compliance with ISO quality standards through dimensional inspection, tensile strength testing and product specification validation.

- Conducted precision measurements using Vernier callipers, micrometer screwgauge and Universal Tensile Strength (UTS) testing equipment to ensure product conformity.
- Identified non-conforming products before downstream processing, reducing the risk of production defects.
- Maintained accurate quality inspection records to support product traceability and continuous quality improvement.

Customer Service Representative– Call Center Operations

Call Center International (CCI)

October 2024 – May 2026

- Resolved customer concerns efficiently through active listening, clear communication, and practical problem solving in a fast-paced environment.
- Maintained high customer satisfaction by balancing empathy with accurate policy interpretation during complex service interactions
- Investigated account issues and guided customers through appropriate solutions reducing repeat contacts and improving overall customer experience.
- Consistently met quality and performance expectations while managing high daily call volumes and adapting to changing operational priorities.
- Collaborated with cross-functional teams to ensure timely issue resolution and seamless customer support.

Laboratory Technician (Attaché)

Materials Testing and Research Division, State Department for Roads

Jan 2023 – Apr 2023

- Conducted laboratory analysis of water, soil, cement and construction materials to support infrastructure quality and regulatory compliance.
- Operated analytical instruments including Atomic Absorption Spectroscopy (AAS) and Flame Photometry to generate accurate laboratory results.
- Prepared technical laboratory reports that informed engineering and construction decisions.
- Maintained strict laboratory quality assurance and safety standards throughout analytical procedures.
- Developed practical experience in chemical testing, laboratory operations and materials research.

Education

Bachelor of Science in Industrial Chemistry

Jomo Kenyatta University of Agriculture and Technology

September 2019 – June 2024

Project

Removal of Pb (II)ions from wastewater using activated carbon prepared from banana peels

Research project at Jomo Kenyatta University of Agriculture and Technology

- Prepared activated carbon from agricultural waste through controlled carbonization and activation processes.
- Evaluated lead removal efficiency using Atomic Absorption Spectroscopy (AAS) by comparing Pb (II)ions concentrations before and after treatment.
- Conducted laboratory experiments involving adsorption studies, sample preparation, centrifugation and analytical testing.
- Explored the potential of sustainable, locally available agricultural waste as a material for wastewater treatment applications.